Automobile Insurance Fraud Detection using Machine Learning

A Systematic Comparison of Resampling Strategies and Supervised versus Unsupervised Approaches

Vichayut Singhakiat
Student ID: 2501013

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Supervisor: Dr. Jianya Lu

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Abstract

Fraud is a costly problem in automobile insurance, and detecting it automatically is challenging primarily because fraudulent claims are rare. In most datasets, genuine claims heavily outnumber fraudulent ones, and this severe class imbalance causes many machine learning models to favor the majority class, consequently missing the critical minority cases. This dissertation examines optimal methods for addressing this imbalance by comparing two distinct detection paradigms on a well-known public dataset of 15,420 automobile (15,419 after cleaning) , of which approximately 6% are fraudulent.

Two primary research questions guided the work. The first investigates how different resampling strategies, namely SMOTE, ADASYN, and a hybrid SMOTE-Tomek method, affect the performance of a supervised gradient boosting classifier. The second compares an unsupervised anomaly detection method (Isolation Forest) with the supervised approach to determine the appropriate contexts for each. To ensure a rigorous comparison, the classifier was held constant while the resampling method was varied, and resampling was exclusively applied to the training data after the train-test split. All models were subsequently evaluated on the same unmodified test set using metrics suitable for imbalanced data, including precision, recall, F1-score, and the area under the precision-recall curve (PR-AUC).

The empirical results demonstrated that resampling provided only marginal gains over a well-configured baseline. Rather than lifting overall performance, resampling primarily shifted the balance between precision and recall; in fact, the baseline model recorded the highest PR-AUC. ADASYN yielded the highest recall, whereas the hybrid SMOTE-Tomek method achieved the best F1-score and the highest precision. Furthermore, the supervised models significantly outperformed the unsupervised Isolation Forest, which successfully flagged most of the fraud but simultaneously generated an impractical number of false positives. The primary conclusion is that, for this dataset, the choice of resampling method is less impactful than commonly assumed, and a supervised gradient boosting model serves as a robust default option. Nevertheless, the unsupervised approach retains practical utility as an early-stage filter or in environments lacking labeled data.

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Chapter 1

Introduction

1.1 Background

Automobile insurance fraud is a serious and costly problem for insurers. It happens when a policyholder makes a dishonest motor claim, for example by inventing an accident that never took place, staging a collision on purpose, or exaggerating the damage or injuries from a real accident (Óskarsdóttir et al., 2022). Motor insurance is especially affected. In the United Kingdom, insurers detected 51,700 fraudulent motor claims worth about £576 million in 2024, which made up 53% of all detected fraudulent insurance claims that year (ABI, 2024). Because these figures only cover fraud that was actually detected, the real scale of motor fraud is likely to be even larger.

Fraudulent motor claims raise costs for insurers, and these extra costs are passed on to honest drivers through higher premiums (ABI, 2024). Detecting fraud quickly and accurately therefore matters to insurers and ordinary customers. For many years, insurers have relied mainly on manual checks by claims investigators. This is slow and very expensive, and it cannot easily keep up with the large number of motor claims made each year. Fraud in car insurance claims is a subject of growing interest, but it remains very difficult to prove. Most fraud is identified by an investigation team, yet such teams cannot realistically check every claim within the time available, and fraudsters constantly adapt their behaviour. As a result, there is growing interest in automated, real-time detection methods that use machine learning to raise potential fraud alerts, so that investigators can focus their limited time on the claims most likely to be fraudulent (Schrijver, Sarmah and El-hajj, 2024; Yankol-Schalck, 2022).

1.2 Problem Statement

Detecting fraud in automobile insurance is difficult mainly because fraud is rare. In the widely used car claims auto insurance dataset, only about 6% of claims are fraudulent (Itri et al., 2020). This severe class imbalance causes many machine learning models to favour the majority, genuine class, so they can look accurate

while still missing most fraud cases (He and Garcia, 2009). Handling this imbalance is therefore the central challenge in auto insurance fraud detection.

A number of studies have tried to solve this problem, and most of them use resampling to rebalance the data. Itri et al. (2020), working on the same car claims dataset, applied SMOTE to oversample the fraud cases and reported that it improved the performance of standard classifiers. Other work has used boosting methods such as XGBoost together with SMOTE and reported high F1-scores (Maina, Moso and Gikunda, 2023). More recent studies have proposed newer hybrid resampling methods for motor insurance, such as combining Center Point SMOTE with random under-sampling, and have tested them across several datasets (Komsrimorakot and Siriborvornratanakul, 2025). This shows that resampling is the standard tool in this field, but it also shows there is no clear agreement on which resampling method is best.

There is a further problem. A systematic literature review by Schrijver, Sarmah and El-hajj (2024), points out that in at least one earlier study oversampling was done before the data was even split into training and testing sets. This can cause data leakage, where information from the test set influences training, making the reported results look better than they really are and impossible to compare fairly with other studies. On top of this, many studies test only a single resampling method, or change several things at once, which makes it hard to see the true effect of the resampling choice by itself.

This project sets out to address these gaps. It carries out a fair, systematic comparison of several resampling methods (SMOTE, ADASYN, and a hybrid SMOTE-Tomek method) while keeping the classifier fixed, so that any difference or impact in performance can be traced to the resampling choice alone. It applies resampling only to the training data, after the train and test split, so the test set keeps its natural 6% fraud rate and prevent the risk of leakage. Finally, it compares this supervised approach against an unsupervised anomaly detection method, the Isolation Forest, to show clearly when each type of method is useful. In these ways the project improves on earlier work by making the comparison fairer and the evaluation more realistic.

1.3 Aim and Objectives

The aim of this dissertation is to build and fairly evaluate machine learning approaches for detecting fraudulent automobile insurance claims, with particular attention to handling severe class imbalance and to comparing supervised and unsupervised methods. The objectives are:

- To review existing research on automobile insurance fraud detection, the class imbalance problem, and resampling and anomaly detection techniques.
- To carry out a systematic comparison of resampling strategies (SMOTE, ADASYN, and SMOTE-Tomek) using a fixed supervised classifier.
- To implement an unsupervised anomaly detection method (Isolation Forest) and compare it with the supervised approach, explaining the assumptions and suitable use of each.
- To evaluate all approaches using metrics suited to imbalanced data, and to interpret the results for real-world auto fraud detection.

1.4 Overview of Methods

This section briefly explains the main methods used in the project. Each one is described in more detail in the Literature Review (Chapter 2) and the Methodology (Chapter 3).

1.4.1 Resampling methods for class imbalance

SMOTE (Synthetic Minority Over-sampling Technique) rebalances the data by creating new, artificial fraud examples instead of simply copying existing ones. For each fraud case, it finds nearby fraud cases and generates a new example somewhere between them (Chawla et al., 2002).

ADASYN (Adaptive Synthetic Sampling) works in a similar way to SMOTE, but it creates more synthetic examples for the fraud cases that are harder to learn, meaning those surrounded by many genuine cases, so that the model pays more attention to the difficult cases (He et al., 2008).

SMOTE-Tomek is a hybrid resampling method. It first applies SMOTE to generate synthetic fraud examples to balance the dataset. Next, it identifies and removes Tomek links, which occur when two data points from opposing classes (one fraudulent claim and one genuine claim) are each other's closest nearest neighbors. Because these paired points sit right next to each other in the data space, they create ambiguity and noise along the decision boundary. Removing them helps clean this boundary, making it easier for the classifier to distinguish between legitimate and fraudulent claims (Batista, Prati and Monard, 2004).

1.4.2 Models

The supervised model is a gradient boosting classifier. Gradient boosting builds many small decision trees one after another, where each new tree tries to correct the mistakes of the previous ones. This kind of model is fast and works well on structured, tabular data such as insurance claims (Ke et al., 2017).

The unsupervised model is an Isolation Forest. Instead of learning from labelled fraud examples, it assumes that fraud is rare and unusual and looks for claims that are easy to separate from the rest of the data. Such claims are treated as anomalies and flagged as possible fraud (Liu, Ting and Zhou, 2008). Its main advantage is that it does not need labelled fraud data, which is often scarce.

1.4.3 This project's approach

The approach taken here is to keep the classifier fixed and change only the resampling method, so the effect of each resampling strategy can be compared fairly. Resampling is applied only to the training data, after the train and test split, to avoid data leakage. The supervised results are then compared with the unsupervised Isolation Forest. All models are evaluated using precision, recall, F1-score, and the area under the precision-recall curve, which are suitable for imbalanced data (Davis and Goadrich, 2006).

1.5 Scope and Limitations

This study focuses only on automobile insurance and uses a single, publicly available labelled dataset. It works with structured, tabular claim data rather than images or written descriptions. The dataset is well known in the research literature, which allows comparison with earlier work, but it is also relatively old; this and other limitations are discussed in Chapter 6.

1.6 Dissertation Structure

The rest of this dissertation is organized as follows. Chapter 2 reviews the literature on auto insurance fraud detection, class imbalance, resampling, and supervised versus unsupervised methods. Chapter 3 explains the methodology in detail. Chapter 4 covers the implementation, Chapter 5 presents the results, Chapter 6 discusses them and their limitations, and Chapter 7 concludes and suggests future work.

Chapter 2

Literature Review

This chapter reviews the research most relevant to the project. It begins with automobile insurance fraud detection and the datasets commonly used in the field. It then discusses the class imbalance problem, resampling techniques, and the difference between supervised and unsupervised approaches. Finally, it explains how performance should be measured on imbalanced data and identifies the gap that this project addresses.

2.1 Automobile Insurance Fraud Detection

Detecting fraud in automobile insurance has been studied for so many years, and machine learning is now the main approach used. Traditional methods relied on fixed rules and manual review by investigators, but these method struggle to cope with the large number of claims and with fraudsters who keep changing their behaviour (Yankol-Schalck, 2022). Machine learning offers a way to learn the patterns of past fraud automatically and to flag suspicious new claims for review.

A systematic literature review by Schrijver, Sarmah and El-hajj (2024) examined many studies on automobile insurance fraud detection using data mining. The review found that a publicly available dataset known as car claims is the most commonly used automobile fraud dataset, appearing in a large number of studies, followed by private datasets obtained directly from insurance companies. This is useful because using the same dataset allows results to be compared across studies. It also means, though, that the field depends heavily on a single, relatively old dataset. The same review warned that some studies report very high accuracy without properly handling imbalance or data leakage, which makes their results hard to trust or compare.

Recent studies continue to use this dataset while trying new techniques. Itri et al. (2020) used SMOTE with several standard classifiers on the carclaims data and studied how different performance measures behave on imbalanced data. More recent work has explored machine learning and new hybrid resampling methods for motor insurance fraud (Komsrimorakot and Siriborvornratanakul, 2025; Yankol-Schalck, 2022). Across all of these studies the same theme keeps appearing: the

main obstacle is the severe imbalance between genuine and fraudulent claims.

2.2 The Class Imbalance Problem

Class imbalance happens when one class has far fewer examples than the other. In fraud detection this is almost always the case, because genuine claims greatly outnumber fraudulent ones. In a widely cited review, He and Garcia (2009) explain that standard machine learning algorithms tend to be biased towards the majority class, because they are designed to reduce overall error. When the minority class is only a small percentage of the data, a model can reach high accuracy while still failing to detect most of the minority cases.

This is why accuracy is not a suitable measure for imbalanced problems. A model that labels every claim as genuine would be about 94% accurate on the carclaims dataset, yet it would catch no fraud at all and be useless in practice. For this reason, researchers use measures that focus on how well the rare class is found, such as precision, recall, and the area under the precision-recall curve (Davis and Goadrich, 2006). There are two broad ways to deal with imbalance: changing the data through resampling, or changing the algorithm, for example by giving more weight to minority-class errors. This project focuses mainly on resampling but also includes a class-weighting model for comparison.

2.3 Resampling Techniques

Resampling changes the balance of the training data. Oversampling adds more minority examples, while undersampling removes majority examples. The simplest form of oversampling is to duplicate existing minority cases, but this can cause the model to overfit, because it sees the same points many times.

SMOTE, introduced by Chawla et al. (2002), avoids this by creating new, synthetic minority examples rather than copies. For each minority case, it finds its nearest minority neighbours and creates a new point somewhere along the line between them. SMOTE is the most widely used oversampling method in imbalanced learning and has inspired many later variants (Fernández et al., 2018). The systematic review by Schrijver, Sarmah and El-hajj (2024) confirmed that SMOTE is by far the most frequently applied oversampling technique in automobile fraud studies. Furthermore, more recent work has proposed improved variants to address this weakness. Bao and Yang (2023), for example, introduced Center Point SMOTE and Inner-Outer SMOTE, which generate synthetic points closer to the centre of the minority distribution to avoid creating noisy samples near the class boundary.

ADASYN, proposed by He et al. (2008), extends this idea. Rather than generating the same number of synthetic points for every minority case, it generates more for the cases that are harder to learn, meaning those surrounded by many majority-class neighbours. The aim is to push the model to focus on the difficult region near the boundary between classes.

Hybrid methods combine oversampling with a cleaning step. SMOTE-Tomek, based on the work of Batista, Prati and Monard (2004), first applies SMOTE and then removes Tomek links, which are pairs of very close points belonging to different classes. Removing these borderline pairs is intended to make the boundary between fraud and genuine claims clearer. Batista, Prati and Monard (2004) found that combining oversampling with cleaning could give better results than oversampling alone.

Hairani, Anggrawan and Priyanto (2023) applied SMOTE-Tomek with a random forest classifier on imbalanced medical data and found that the hybrid method improved accuracy, precision, and F1-score compared with using SMOTE alone, supporting the value of combining oversampling with a cleaning step.

It is worth noting that resampling is not always helpful. Elor and Averbuch-Elor (2022) showed that, with strong modern classifiers, the benefit of SMOTE and similar methods can be small or even disappear. This makes a careful, systematic comparison, rather than assuming one method is best, especially valuable, and it is one of the main reasons this project compares several resampling methods under the same conditions.

2.4 Supervised and Unsupervised Approaches

Fraud detection can follow two different paradigms. The supervised approach learns from labelled examples: it is told which past claims were fraudulent and which were genuine, and it learns the patterns that separate them. Gradient boosting methods, such as the LightGBM model described by Ke et al. (2017), are among the strongest supervised classifiers for structured, tabular data. They build many small decision trees in sequence, with each tree correcting the errors of the previous ones, and they are fast enough to handle large datasets.

A practical example of this behaviour is reported by Permai and Herdianto (2023), who used logistic regression and XGBoost to detect fraud in insurance claims. Their XGBoost model reached about 85% accuracy but a recall of only around 23%, which shows that a high accuracy can hide poor detection of the minority class and confirms why accuracy alone is not a suitable measure in this setting.

The unsupervised approach does not use labels at all. Instead, it assumes that fraud is rare and unusual, and it searches for claims that do not fit the normal

pattern. Isolation Forest, introduced by Liu, Ting and Zhou (2008), is a popular method of this type. It works by randomly splitting the data many times; points that can be separated from the rest with only a few splits are treated as anomalies. Its main advantage is that it does not need labelled fraud examples, which are often scarce or costly to obtain. Its main weakness is that not every unusual claim is fraudulent, so it can produce many false alarms.

The two methods make different assumptions and suit different situations. Supervised methods are appropriate when there is a good amount of labelled data and when future fraud is expected to resemble past fraud. Unsupervised methods are more appropriate when labels are unavailable, or when new and previously unseen types of fraud must be detected. Comparing the two, as this project does, helps to show these trade-offs clearly and to give practical guidance on when each should be used.

2.5 Measuring Performance on Imbalanced Data

As noted above, accuracy is misleading on imbalanced data. This project therefore uses precision, recall, and F1-score, together with the area under the precision-recall curve (PR-AUC). Precision measures how many of the claims flagged as fraud really are fraud, while recall measures how many of the actual fraud cases were found. There is usually a trade-off between the two. Davis and Goadrich (2006) argue that precision-recall curves give a more informative picture than ROC curves when the positive class is rare, because ROC curves can look overly optimistic in these conditions. For this reason, PR-AUC is treated as the main summary measure in this dissertation, with ROC-AUC reported alongside it for completeness.

2.6 Research Gap

The literature shows that class imbalance is the central challenge in automobile insurance fraud detection, that many resampling methods exist, and that supervised and unsupervised approaches each have strengths and weaknesses. Three weaknesses appear repeatedly, though. First, many studies test only one resampling method, or change the model and the resampling at the same time, which makes it hard to see the true effect of the resampling choice. Second, at least one study applied resampling before splitting the data, which risks data leakage and unrealistic results (Schrijver, Sarmah and El-hajj, 2024). Third, accuracy is still sometimes reported instead of imbalance-aware metrics. This project addresses these gaps by holding the classifier fixed while systematically varying the resampling method, by resampling

only the training data after the split, by including an unsupervised baseline for contrast, and by evaluating everything with imbalance-aware metrics on an untouched test set.

Chapter 3

Methodology

This chapter sets out how the study was carried out. It describes the dataset, the cleaning and encoding steps, the way the data was split to avoid leakage, the resampling methods, the exact model settings, and the evaluation metrics. The design follows the plan described in Chapters 1 and 2: the classifier is kept fixed while the resampling method is changed, so that any difference in results can be linked to the resampling choice specifically rather than to the model.

3.1 Research Design

The study is an experimental comparison. Six conditions were run in total. Five of them use the same supervised classifier with different treatments of the imbalance (no resampling, class weighting, SMOTE, ADASYN, and SMOTE-Tomek), and the sixth is an unsupervised Isolation Forest. Because only the treatment of the data changes between the five supervised runs, the effect of each resampling method can be judged fairly. The Isolation Forest is included as a contrasting approach that does not use labels at all.

3.2 Dataset

The study uses the publicly available car claims automobile insurance dataset, downloaded from Kaggle (Kaggle, 2021). As downloaded, it contains 15,420 claims, each described by 32 features plus a binary label showing whether the claim was found to be fraudulent. The features describe the policyholder, the vehicle, the accident, the claim, and the insurance policy. After removing one invalid record during data cleaning (see Section 3.3), 15,419 claims remain. Of these, 14,496 (about 94%) are genuine and 923 (about 6%) are fraudulent, which confirms the severe class imbalance discussed in Chapter 2. The same "carclaims" automobile dataset has been used in other recent studies, including Özaltın and Karadağ Erdemir (2025), which allows the results of this project to be compared with published work.

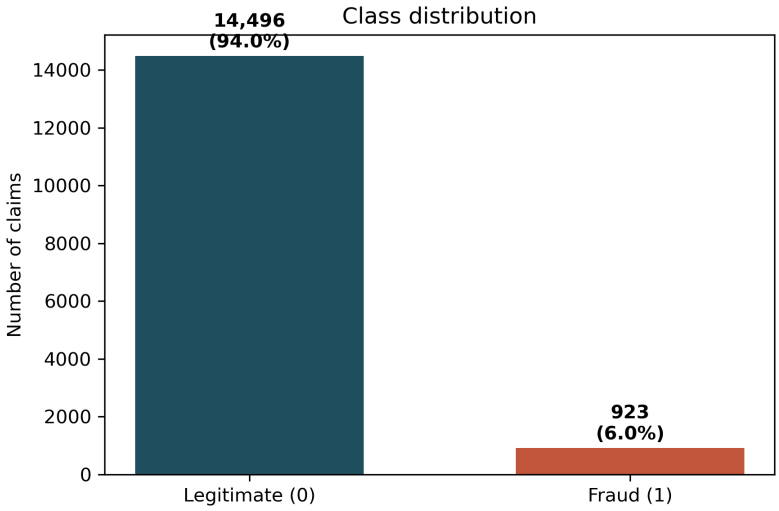


Figure 3.1: Class distribution of the dataset. Fraudulent claims make up only about 6% of the data.

A short exploratory analysis showed that some features are clearly related to fraud. For example, the fraud rate is much higher when the policyholder is at fault (about 7.9%) than when a third party is at fault (about 0.9%), and it is higher for All Perils policies than for Liability policies. This suggests the features carry useful signal, even though fraud is rare overall.



Figure 3.2: Fraud rate by selected features. The dashed line shows the overall fraud rate of about 6%.

3.3 Data Cleaning

The dataset had no missing values and no duplicate rows, but a few invalid entries were found and dealt with. Some 320 records had an age of 0, which is not possible for a policyholder, so these were replaced with the median age of the valid records. One record had an invalid value of 0 for the claimed month, so this single row was

removed, leaving 15,419 claims. The policy number was also removed, because it is only an identifier and carries no predictive power.

3.4 Feature Encoding

The dataset is mostly made up of categorical features, with only a few numerical ones (after cleaning, 24 categorical and 7 numerical). Because the models used in this study are tree based, the categorical features were converted to integer codes using ordinal encoding. Tree based models can work with these integer codes directly, so more complex encoding such as one-hot encoding was not required. Avoiding one-hot encoding also keeps the number of features from growing very large, which can happen when categories have many possible values.

3.5 Train and Test Split, and Avoiding Data Leakage

A central part of the design is to avoid data leakage, which is one of the most common and most serious mistakes in this kind of study, and one that Chapter 2 identified in earlier work. Data leakage would occur if information from the test set influenced the training process, for example if synthetic examples created by resampling appeared in, or closely resembled, the test data. This would make the results look far better than they really are.

Kaufman et al. (2012) give a formal treatment of leakage in data mining, defining it as the introduction of information about the target that should not legitimately be available at prediction time, and note that it is one of the most common causes of inflated results. The scale of the problem is not trivial: a survey by Kapoor and Narayanan (2023) identified leakage across 17 scientific fields, affecting 294 papers, in some cases producing substantially over-optimistic conclusions. Applying resampling before the split is a textbook instance of this failure, because synthetic minority points generated from training cases can closely resemble, or be interpolated towards, points that end up in the test set.

To prevent this, the data was split into training and test sets before any resampling was applied. A stratified split was used, keeping 75% of the data for training and 25% for testing, with the same 6% fraud rate in both parts. A fixed random seed of 42 was used so the split can be reproduced. Resampling was then applied only to the training set. The test set was never resampled and was kept aside until the final evaluation, so it always reflects the natural, real-world balance of the data.

Table 3.1: The stratified train and test split. Only the training set is resampled; the test set keeps its natural balance.

Set	Rows	Fraud rate	Resampled?
Training set	11,564	about 6%	Yes (training only)
Test set	3,855	about 6%	No (kept untouched)

3.6 Resampling Methods Compared

Five training conditions were compared for the supervised model, all using the same classifier so that differences come from the resampling choice:

1. **Baseline:** No resampling (the classifier is trained on the original imbalanced training data).

2. **Class Weights:** No resampling is applied, but the classifier gives more importance to fraud cases by weighting them in inverse proportion to their frequency.

3. **SMOTE:** Synthetic fraud examples are added to the training set until the two classes are balanced (Chawla et al., 2002).

4. **ADASYN:** Similar to SMOTE, but more synthetic examples are created for the harder-to-learn fraud cases (He et al., 2008).

5. **SMOTE-Tomek:** SMOTE is applied first, then borderline pairs from different classes (Tomek links) are removed to clean the boundary (Batista, Prati and Monard, 2004).

Resampling applies only to the supervised classifier. The unsupervised Isolation Forest does not use labels during training, so there is no minority class to oversample and resampling does not apply to it. This is not a flaw in the design but a genuine difference between the two paradigms, and it reflects how they would be used in practice.

3.6.1 How the Resampling Methods Work

Because the resampling methods are central to this project, this section explains how each one actually works and how they differ from one another. All three are oversampling methods, which means they add new fraud examples rather than removing genuine ones, but they go about it in different ways.

Random oversampling (the simplest idea)

The most basic way to balance the data is to copy existing fraud cases at random until there are as many fraud cases as genuine ones. The problem is that the model then sees the very same fraud rows many times, which can lead it to memorize those specific cases rather than learn the general pattern of fraud. This is called overfitting. The methods below were designed to avoid this weakness.

SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) does not copy fraud cases. Instead, it creates new, artificial fraud examples that are similar to the real ones but not identical (Chawla et al., 2002). For each fraud case, it finds its nearest fraud neighbours in the data. It then picks one of those neighbours and creates a brand-new point somewhere on the straight line between the original case and the neighbour. In simple terms, if two real fraud claims are alike, SMOTE invents a new claim that sits between them and shares their features. This fills in the space around the fraud cases and gives the model more variety to learn from, which reduces overfitting compared with plain copying.

ADASYN

ADASYN (Adaptive Synthetic Sampling) works in almost the same way as SMOTE, with one key difference: it does not treat every fraud case equally (He et al., 2008). Before creating new points, it looks at each fraud case and checks how many of its neighbors are genuine claims. Fraud cases surrounded by many genuine claims are considered harder to learn, because they sit close to the boundary between the two classes. ADASYN then creates more synthetic examples around these hard cases and fewer around the easy ones. The idea is to push the model to pay extra attention to the difficult region where fraud and genuine claims overlap. A side effect, as seen later, is that ADASYN does not always produce a perfectly equal balance, because the number of new points depends on how hard each case is.

SMOTE-Tomek

SMOTE-Tomek is a hybrid method that combines oversampling with a cleaning step (Batista, Prati and Monard, 2004). It first runs SMOTE to add synthetic fraud examples, exactly as described above. It then searches for Tomek links, which are pairs of claims that are each other's nearest neighbour but belong to different classes, meaning one is fraud and the other is genuine and they sit right next to each other. These pairs mark the messy, overlapping region between the classes.

SMOTE-Tomek removes the genuine claim from each such pair, which cleans up the boundary and makes the line between fraud and genuine a little clearer. The result is a balanced dataset like SMOTE, but with some of the most confusing borderline points tidied away.

3.7 Models and Settings

3.7.1 Supervised model

The supervised model is a gradient boosting classifier from the LightGBM family (Ke et al., 2017), which builds decision trees in sequence and groups continuous values into bins for fast training. Gradient boosting is well suited to structured, tabular data of this kind and is a strong, widely used baseline in fraud detection research. The same settings were used in every supervised condition, so that only the training data changed between experiments. Keeping the model fixed is what makes the comparison of resampling methods fair. The main settings were:

- Number of trees (n_estimators): 300
- Learning rate: 0.05
- Maximum tree depth: 6
- Number of leaves: 31
- L2 regularisation (reg_lambda): 1.0
- Random seed: 42

These values give a moderately sized, regularized model that trains quickly and is not too likely to overfit. The settings were kept fixed rather than heavily tuned, because the goal of the study is to compare resampling methods on equal terms, not to squeeze out the best possible score from one model.

3.7.2 Unsupervised model

The unsupervised model is an Isolation Forest (Liu, Ting and Zhou, 2008) with 300 trees and a random seed of 42. The contamination setting, which tells the model roughly how much of the data is expected to be anomalous, was set to match the fraud rate in the training data (about 6%). The model was trained only on genuine claims and then used to score test claims by how unusual they are, with more unusual claims treated as more likely to be fraud. It does not use fraud labels during training, which is exactly the setting in which an unsupervised method would be chosen in practice.

3.7.3 How the Models Work

This section briefly explains the theory behind the two types of model used, so that the later results can be understood properly.

Gradient boosting was formalised by Friedman (2001) as a stage-wise additive model in which each new learner is fitted to the gradient of the loss function of the current ensemble. Gradient boosting, used here through LightGBM, is a supervised method that builds a strong model out of many weak ones (Ke et al., 2017). It creates a sequence of small decision trees, where each new tree focuses on the mistakes made by the trees before it. A decision tree is simply a set of yes or no questions about the features, such as whether the claim was the policyholder's fault, that lead to a prediction. On its own a small tree is weak, but by adding hundreds of them together, each correcting the last, gradient boosting produces an accurate overall model. LightGBM speeds this up by grouping continuous values into bins, which is why it works well on large tabular datasets. Because it learns from the fraud labels, it needs a reasonable number of labelled fraud examples to work well, which is where resampling helps.

Isolation Forest is an unsupervised method, meaning it does not use the fraud labels at all (Liu, Ting and Zhou, 2008). It is based on a simple idea: unusual points are easier to separate from the rest of the data than normal points. The method builds many trees by repeatedly splitting the data at random. A claim that gets separated from the others after only a few splits is considered unusual, and therefore a possible anomaly or fraud. Because it looks for unusual points rather than learning what fraud looks like, it does not need labelled data, but it also cannot tell the difference between fraud and a claim that is simply unusual for an innocent reason. This is the key theoretical reason for its later behaviour, where it flagged many genuine claims as suspicious.

3.8 Evaluation Metrics

Because the data is imbalanced, accuracy was not used as a main measure. Instead the following metrics were reported, all calculated on the untouched test set:

- Precision: of the claims flagged as fraud, how many were really fraud.
- Recall: of all the real fraud cases, how many were found.
- F1-score: the harmonic mean of precision and recall, giving a single balanced measure.

- PR-AUC (area under the precision-recall curve): the main summary measure, suited to rare positive classes (Davis and Goadrich, 2006).
- ROC-AUC: reported alongside for completeness, though it is less informative than PR-AUC when the positive class is rare.

Precision, recall, and F1-score are calculated from the confusion matrix as follows (Hairani, Anggrawan and Priyanto, 2023):

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{3.1}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3.2}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.3}$$

where TP is the number of fraudulent claims correctly identified, FP is the number of genuine claims wrongly flagged as fraud, and FN is the number of fraudulent claims that were missed.

For the metrics that need a decision threshold (precision, recall, and F1), the threshold was chosen to give the best F1-score on the precision-recall curve, so that each model is judged at its own best balance point. PR-AUC and ROC-AUC do not depend on a single threshold, as they summarise performance across all thresholds.

3.9 Tools

The analysis was carried out in Python. The pandas (McKinney, 2010) and NumPy (Harris et al., 2020) libraries were used for data handling, scikit-learn (Pedregosa et al., 2011) for the train and test split, encoding, Isolation Forest, and evaluation metrics, the LightGBM library (Ke et al., 2017) for the gradient boosting classifier, the imbalanced-learn library (Lemaître, Nogueira and Aridas, 2017) for SMOTE, ADASYN, and SMOTE-Tomek, and Matplotlib (Hunter, 2007) for the figures. All work was done in a Jupyter notebook so that each step can be run and checked in order.

Chapter 4

Implementation

This chapter shows how the methodology in Chapter 3 was put into code. The most important parts are shown below with short explanations. The complete code is provided as a Jupyter notebook that can be run from top to bottom.

4.1 Loading and Cleaning the Data

The data was first loaded and cleaned. The policy number was dropped, impossible ages were replaced with the median age, and the single invalid row was removed. The categorical features were then converted to integer codes using ordinal encoding.

```
import pandas as pd, numpy as np
from sklearn.preprocessing import OrdinalEncoder

df = pd.read_csv('fraud_oracle.csv')
df = df.drop(columns=['PolicyNumber'])
RANDOM_STATE = 42
np.random.seed(RANDOM_STATE)

# replace impossible Age == 0 with the median of valid ages
median_age = int(df.loc[df['Age'] > 0, 'Age'].median())
df.loc[df['Age'] == 0, 'Age'] = median_age

# remove the single invalid row (MonthClaimed == '0')
before = len(df)
df = df[df['MonthClaimed'] !='0'].reset_index(drop=True)

y = df['FraudFound_P'].values
X = df.drop(columns=['FraudFound_P'])
cat_cols = X.select_dtypes(include='object').columns.tolist()
num_cols = X.select_dtypes(include=['int64', 'float64']).
   columns.tolist()
encoder = OrdinalEncoder(handle_unknown='use_encoded_value',
   unknown_value=-1)
```

```
X_enc = X.copy()
X_enc[cat_cols] = encoder.fit_transform(X[cat_cols])
X_enc = X_enc.astype(float).values
```

4.2 Splitting Before Resampling

The data was split into training and test sets before any resampling, using a stratified split so both sets keep the same fraud rate. This is the step that prevents data leakage.

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X_enc, y, test_size=0.25,
    stratify=y, random_state=RANDOM_STATE)
# resampling is applied ONLY to X_train / y_train
# X_test / y_test are never resampled
```

4.3 Applying the Resampling Methods

The resampling methods come from the imbalanced-learn library. Each one is fitted on the training data only. The example below shows SMOTE; ADASYN and SMOTE-Tomek are used in the same way.

```
from imblearn.over_sampling import SMOTE, ADASYN
from imblearn.combine import SMOTETomek

# SMOTE (ADASYN and SMOTETomek follow the same pattern)
X_res, y_res = SMOTE(random_state=RANDOM_STATE).fit_resample(
    X_train, y_train)
print(np.bincount(y_res))   # classes now balanced
```

4.4 Training and Evaluating the Models

The same LightGBM model was trained under each resampling condition, and the Isolation Forest was trained on genuine claims only. Every model was then evaluated on the untouched test set. The threshold for precision, recall, and F1 was chosen to give the best F1 on the precision-recall curve.

```
from lightgbm import LGBMClassifier
from sklearn.ensemble import IsolationForest
from sklearn.metrics import (average_precision_score,
    precision_recall_curve, f1_score)

def make_model():
    return LGBMClassifier(n_estimators=300, learning_rate
       =0.05,
        max_depth=6, num_leaves=31, reg_lambda=1.0,
        random_state=RANDOM_STATE, verbose=-1)

# supervised example (SMOTE condition)
model = make_model().fit(X_res, y_res)
scores = model.predict_proba(X_test)[:, 1]
pr_auc = average_precision_score(y_test, scores)

# unsupervised Isolation Forest
iso = IsolationForest(n_estimators=300,
        contamination=y_train.mean(), random_state=42)
iso.fit(X_train[y_train == 0])       # genuine claims only
iso_scores = -iso.score_samples(X_test)
```

Chapter 5

Results

This chapter presents the results of the six experiments. All figures are calculated on the untouched test set of 3,855 claims. The main results are summarised in Table 5.2, followed by the precision-recall curves and confusion matrices.

5.1 Effect of Resampling on the Training Data

Before looking at model performance, it is useful to see how each resampling method actually changed the training data. This gives a concrete picture of what the methods did, rather than only describing them in theory. The training set began with 10,872 genuine claims and only 692 fraud claims, a ratio of about 1 fraud to every 16 genuine claims. Table 5.2 and Figure 5.1 show the counts after each method was applied.

Table 5.1: Training set class counts before and after each resampling method. The test set was never resampled.

Condition	**Genuine (0)**	**Fraud (1)**	**Total**	**Balance**
Before (original)	10,872	692	11,564	1: 15.7
After SMOTE	10,872	10,872	21,744	1: 1.0
After ADASYN	10,872	10,960	21,832	1: 0.99
After SMOTE-Tomek	10,870	10,870	21,740	1: 1.0

Several things stand out. SMOTE brought the fraud class up from 692 to 10,872, exactly matching the genuine class, so the two are perfectly balanced. SMOTE-Tomek produced almost the same result, but the Tomek cleaning step removed two genuine claims that were sitting right on the boundary, which is why the genuine count dropped very slightly to 10,870. ADASYN behaved a little differently: it created slightly more fraud examples than genuine ones (10,960 against 10,872), because it added extra synthetic points around the hard-to-learn fraud cases, and the totals depend on how many hard cases it found rather than on a fixed target. Although this difference is small here, it reflects the adaptive nature of ADASYN and helps explain why it later produced the highest recall, as discussed in Chapter 6.

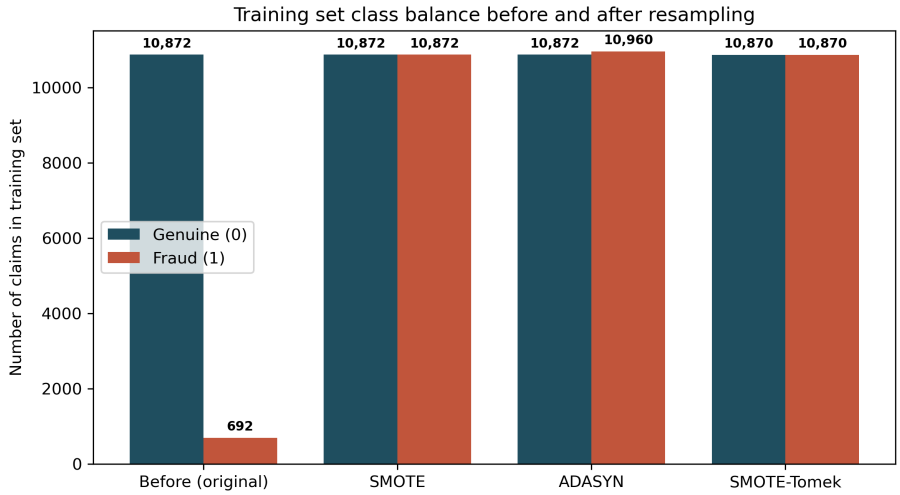


Figure 5.1: Class balance in the training set before and after resampling. All three methods bring the two classes to roughly equal size.

5.2 Overall Comparison

Table 5.2: Performance of all models on the untouched test set. The best supervised F1 and precision (SMOTE-Tomek) are highlighted. Isolation Forest reaches high recall but very low precision.

Model	Precision	Recall	F1	PR-AUC	ROC-AUC
Baseline (no resampling)	0.208	0.502	0.294	**0.252**	**0.839**
Class weights	0.226	0.429	0.296	0.227	0.833
SMOTE	0.230	0.437	0.301	0.240	0.836
ADASYN	0.203	0.511	0.291	0.235	0.834
SMOTE-Tomek (hybrid)	**0.233**	**0.468**	**0.311**	0.243	0.837
Isolation Forest	0.064	0.710	0.117	0.068	0.513

Among the supervised models the differences are small. The baseline already performs reasonably well, with a PR-AUC of 0.252 and a ROC-AUC of 0.839. Adding resampling changes recall and precision slightly but does not transform the results. ADASYN gives the highest recall (0.511), meaning it finds the largest share of true fraud cases, while the hybrid SMOTE-Tomek method gives the highest F1-score (0.311) and the highest precision (0.233). Interestingly, the plain baseline reaches the highest PR-AUC and ROC-AUC values, which suggests that resampling did not improve the overall ranking of claims by fraud risk, even though it shifted the balance between precision and recall. SMOTE gave the second highest F1 (0.301), while class weighting performed at a similar level to the baseline.

The unsupervised Isolation Forest behaves very differently. It reaches a high recall of 0.710, finding most of the fraud cases, but its precision is only 0.064,

meaning that the large majority of the claims it flags are actually genuine. Its PR-AUC of 0.068 is close to the level expected from random guessing on this data, and its ROC-AUC of 0.513 is only just above 0.5. This shows that, on its own, the unsupervised method cannot separate fraud from genuine claims nearly as well as the supervised models.

5.3 Supervised Metric Comparison

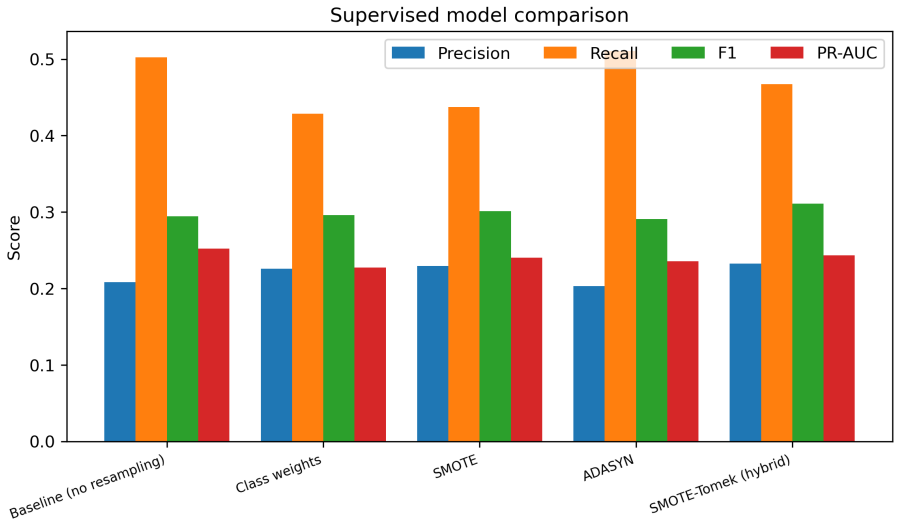


Figure 5.2: Precision, recall, F1, and PR-AUC for the supervised models. Differences between resampling methods are small; ADASYN favours recall, while SMOTE-Tomek gives the best F1 and precision.

5.4 Precision-Recall Curves

The precision-recall curves in Figure 5.3 show performance across all possible thresholds, rather than at a single point. The supervised models cluster closely together and sit well above the level expected from random guessing, while the Isolation Forest curve stays very low. This confirms the pattern seen in Table 5.2: the resampling methods make only a small difference, but the gap between the supervised and unsupervised approaches is large.

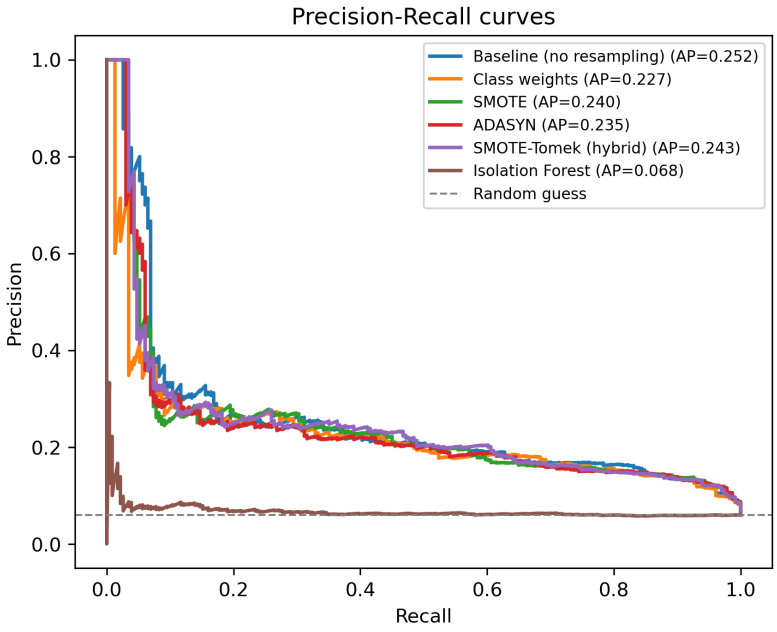


Figure 5.3: Precision-recall curves for all models. The dashed line marks the fraud rate expected from random guessing.

5.5 Confusion Matrices

Figure 5.4 compares the confusion matrices of the best supervised model (SMOTE-Tomek) and the Isolation Forest, each at its best-F1 threshold. The difference in behaviour is clear. The supervised model correctly identified 108 of the 231 fraud cases in the test set while raising 356 false alarms. The Isolation Forest caught more fraud (164 cases) but raised 2,408 false alarms, which would be far too many for investigators to handle in practice.

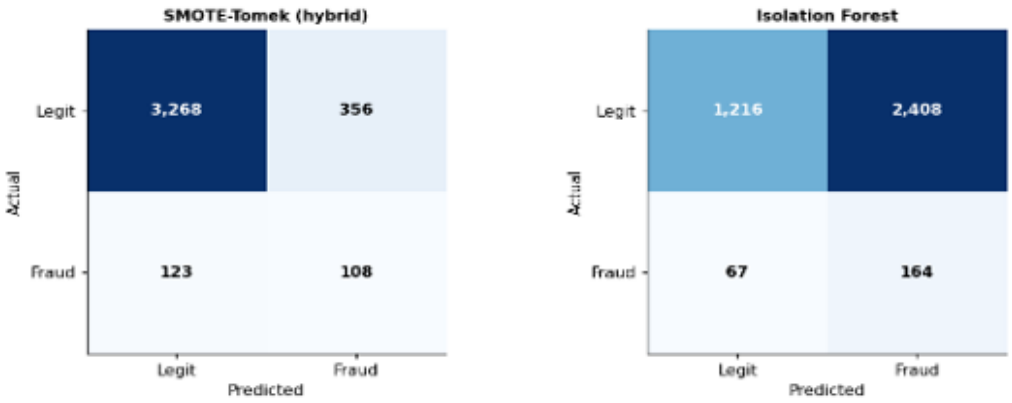


Figure 5.4: Confusion matrices for the best supervised model (left) and the Isolation Forest (right). The unsupervised model produces far more false positives.

Chapter 6

Discussion

This chapter interprets the results in relation to the aims of the study, and then considers the limitations of the work.

6.1 The Effect of Resampling

The first aim was to see how SMOTE, ADASYN, and SMOTE-Tomek affect the performance of the supervised classifier. The results show that their effect is real but modest. Compared with the baseline, the resampling methods shifted the balance between precision and recall, but they did not clearly improve the overall summary measure (PR-AUC), and the baseline in fact recorded the highest PR-AUC of all. ADASYN gave the strongest recall (0.511), which fits its design of focusing on harder minority cases, while SMOTE-Tomek gave the best balance of precision and recall as measured by F1 (0.311), with SMOTE close behind (0.301).

It is also worth noting that no single method was best on every measure. The baseline had the best PR-AUC and ROC-AUC, SMOTE-Tomek the best F1 and precision, and ADASYN the best recall. This spread supports the view of Elor and Averbuch-Elor (2022) that, with a strong modern classifier, resampling may add less overall value than is often assumed, and that its main effect is to move the precision-recall balance around rather than to lift performance as a whole. In practical terms, this suggests that an insurer using a gradient boosting model should choose a resampling method (or none at all) based on the precision-recall balance it wants, rather than expecting resampling to give a large across-the-board gain. It also echoes the wider point from the literature that high headline results reported in some earlier studies may owe as much to their evaluation setup as to the resampling method itself.

6.2 Supervised versus Unsupervised

The second aim was to compare the unsupervised Isolation Forest with the supervised approach and to work out when each is more suitable. The results give a clear answer for this dataset: the supervised approach is far more effective. The Isolation

Forest reached high recall but extremely low precision, and its overall PR-AUC was close to random. This happens because it treats any unusual claim as suspicious, and many genuine claims are also unusual in some way.

This does not make the unsupervised approach worthless. Its main strength is that it does not need labelled fraud examples, which are often scarce. In a setting where no labels are available, or where completely new types of fraud must be detected, an unsupervised method could still work as an early filter that reduces the number of claims a supervised system or a human has to examine. The key point is that supervised methods are the better choice when good labels exist and future fraud resembles past fraud, while unsupervised methods are mainly valuable when labels are missing or novel fraud is a concern.

6.3 Precision, Recall, and Practical Meaning

A third theme running through the results is the trade-off between precision and recall, and what it means in practice. In fraud detection, false positives and false negatives carry different costs. A false positive, where a genuine claim is flagged, wastes investigator time and may annoy an honest customer. A false negative, where a fraud is missed, means the insurer pays out on a bogus claim. The best threshold therefore depends on how an insurer values these two kinds of error.

The results show that the supervised models let this balance be tuned sensibly, because their precision stays at a useful level as recall increases. The Isolation Forest, by contrast, only offers high recall at a precision so low that most flagged claims are genuine, which would overwhelm investigators. For a real deployment, a supervised model with a threshold chosen to match the insurer's capacity for investigation would be the most practical option. Where recall matters most, ADASYN or the baseline would be preferred, since they found the most fraud; where a balance of precision and recall is wanted, SMOTE-Tomek would be the sensible choice, as it gave the best F1 and the highest precision.

6.4 Why the Methods Behaved as They Did

It is worth going beyond the numbers to explain why the different resampling methods led to the differences seen in the results. The explanations below link each method's design, described in Chapter 3, to its behaviour on the test set.

ADASYN produced the highest recall among the resampling methods. This follows directly from how it works. ADASYN adds more synthetic fraud examples in the regions where fraud cases are surrounded by genuine claims, which are the

hard, overlapping areas near the boundary between the two classes. As shown in Chapter 5, it also produced slightly more fraud examples than the other methods. By flooding these difficult regions with extra fraud points, ADASYN pushes the model to treat borderline claims as fraud more readily. This raises recall, because more real fraud cases are caught, but it also lowers precision, because more genuine claims that happen to sit near the boundary are wrongly flagged. The result seen in Chapter 5, where ADASYN had the highest recall but the lowest precision of the supervised models, is exactly what its design would predict.

SMOTE-Tomek produced the best F1 and the highest precision. Again this matches its design. SMOTE-Tomek adds synthetic fraud points like SMOTE, but then removes Tomek links, which are the confusing pairs of opposite-class claims sitting right on the boundary. Cleaning away these borderline points makes the line between fraud and genuine a little sharper, so the model makes fewer mistakes on claims near the boundary. This improves precision, because fewer genuine claims are wrongly flagged, and it gives the best overall balance as measured by F1. The trade-off is slightly lower recall than ADASYN, because the sharper boundary makes the model a little more cautious about calling a claim fraud.

Plain SMOTE sat between the two, which is sensible because it neither concentrates on the hard cases like ADASYN nor cleans the boundary like SMOTE-Tomek. It spreads new fraud points evenly, so it gave a moderate, balanced result without a strong tilt towards either recall or precision.

Perhaps the most important observation is why resampling helped so little overall, with the plain baseline even recording the best PR-AUC. The likely reason is that gradient boosting is already a strong learner that can handle a fair amount of imbalance on its own, partly because it builds trees in sequence that focus on the cases it is getting wrong, including the rare fraud cases. When the model is already this capable, adding synthetic data mainly moves the precision-recall balance around rather than giving the model genuinely new information. This is consistent with the findings of Elor and Averbuch-Elor (2022), who showed that the benefit of SMOTE-style methods can be small or absent with strong modern classifiers. In other words, the resampling methods changed how the model traded off precision against recall, but they did not improve its underlying ability to rank claims by fraud risk, which is what PR-AUC measures.

The unsupervised Isolation Forest behaved very differently for a deeper reason than simply being weaker. It does not learn what fraud looks like; it only looks for claims that are unusual. In insurance data, many perfectly genuine claims are unusual in some way, for example a rare vehicle type or an uncommon combination of policy features. The Isolation Forest cannot tell an unusual-but-genuine claim from an unusual-and-fraudulent one, so it flags both. This is why it caught most fraud

(high recall) but buried it among thousands of false alarms (very low precision), and why its ranking of claims by risk was close to random. The problem is not the amount of data but the fundamental limit of an approach that equates unusual with suspicious.

6.5 Limitations

Several limitations should be kept in mind when reading these results.

First, the dataset, although widely used, is relatively old, and fraud patterns, vehicles, and claims processes have changed since it was collected, which limits how far the results apply to current practice.

Second, only detected fraud is labelled, so some fraud in the data may be unlabelled, which could affect both training and evaluation.

Third, the study used a single dataset, so the findings may not carry over to other insurers or types of insurance.

Fourth, the study used a single train and test split rather than repeated splits or cross-validation, so the exact figures could vary a little with a different split. Kohavi (1995) showed that single hold-out estimates carry higher variance than repeated resampling schemes such as stratified ten-fold cross-validation. The small differences between resampling methods should therefore be read as indicative rather than definitive, although the fixed random seed makes the figures fully reproducible.

Fifth, the decision threshold for precision, recall, and F1 was selected to maximise F1 on the test set, so these three figures are optimistically biased. Because the same procedure was applied identically to every model, the comparison between conditions remains fair, and the headline measures used in this study, PR-AUC and ROC-AUC, are threshold-free and therefore unaffected. A stricter design would select the threshold on a separate validation split.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This dissertation set out to build and fairly evaluate machine learning approaches for detecting fraudulent automobile insurance claims, with a focus on handling severe class imbalance and on comparing supervised and unsupervised methods. Using a well-known public dataset of 15,420 claims (15,419 after cleaning) with a 6% fraud rate, it carried out a systematic comparison in which the classifier was held fixed while the resampling method was varied, and it added an unsupervised Isolation Forest for contrast. Throughout, resampling was applied only to the training data after the split, so the evaluation was carried out on realistic, untouched test data.

Three main conclusions come out of the work. First, resampling had only a modest effect: SMOTE, ADASYN, and SMOTE-Tomek shifted the balance between precision and recall but did not clearly improve the overall ranking of claims, and the baseline model actually recorded the highest PR-AUC. Second, the supervised approach was far more effective than the unsupervised one, which found a lot of fraud but flagged far too many genuine claims to be useful on its own. Third, the best method in practice depends on the balance an insurer wants between catching fraud and avoiding false alarms, and on whether labelled data is available. Overall, the study suggests that, for this kind of data, a supervised gradient boosting model is a dependable choice, and that the resampling method matters less than is sometimes assumed, mainly changing the precision-recall balance rather than lifting performance overall.

7.2 Future Work

Several directions could build on this work. Testing the same design on newer and larger datasets, ideally from more than one insurer, would show how well the findings generalise. Using repeated splits or cross-validation would give a clearer sense of how stable the results are. Comparing more resampling methods, including other SMOTE variants and undersampling techniques, would give a fuller picture. Adding explainability tools such as SHAP (Lundberg and Lee, 2017) would help

investigators understand why individual claims are flagged, which matters for real use. Finally, combining the two approaches, for example using an unsupervised filter followed by a supervised classifier, could be explored as a way to get the benefits of both.

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Appendix

Dataset Source

The automobile insurance fraud dataset used in this study is publicly available on Kaggle. It can be accessed at the following link: https://www.kaggle.com/datasets/shivamb/vehicle-claim-fraud-detection/data

Jupyter Notebook Code

The complete implementation of this project, including data cleaning, resampling strategies, and model evaluation, was carried out using Python in a Jupyter Notebook. The full notebook containing all the code is provided as a separately submitted PDF document.